

STEPCLIMBER · DESIGN DECISION RECORD

StepClimber recommendations accepted

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DOCUMENT CONTROL

Document ID	SCM-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepclimber		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Items 9, 10 and 14 decided by Amish on 2026-10-02 as recommended in SCM-DEC-001
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); record the decided items, the changes made at TRL 3 and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items 11 to 13 of SCM-DDR-001); items 9, 10 and the new item 14 were decided by Amish on 2026-10-02 as recommended in the design decisions register (SCM-DEC-001, items 6, 7 and 4): "i approve your recommendations for all 555 open decisions."

■ Context

SCM-DDR-001 left five StepClimber items open. Three of them carried a recommendation: the cluster and drive rework for R2 and R15 (item 11), the tilt window for R6 (item 12) and the follow-on targets for mass, budget and speed if the rework went ahead (item 13). On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Each item with a recommendation is therefore **Decided by Amish, 2026-09-25: go with recommendation**. Items with no recommendation stay open. TRL 4 remains on hold by Amish's instruction, so everything below was carried out on paper at TRL 3.

■ Decision

Table 1. Newly decided items and what changed in the repo.

#	Item	Decision	What changed
11	Cluster and drive rework (R2, R15)	Option A: 150 mm arms (were 135 mm), 200 mm wheels (were 150 mm) and a two-stage chain drive with an 08B 20-tooth final sprocket on the cluster shaft	<code>cad/src/model.py</code> : new cluster, a 20 mm countershaft 150 mm up the frame, first stage 06B 10T to 35T, final stage 08B 10T to 20T (7:1 overall, was 6:1), rail extended so the toe plate stays about 25 mm off the floor; STEP and STL re-exported. SCM-DWG-001 moved to Rev P2. BOM items 2 and 3 repriced (\$65 to \$88; \$50 to \$80). SCM-CAL-001 v0.2: shaft-line guard radius 53.9 mm inside the 54.2 mm envelope (was 104 mm against 48 mm), arms clear 32 mm nosing overhangs (was 23 mm). R2 and R15 now met
12	Tilt window (R6, R7)	Tighten the stop window from plus or minus 8 to plus or minus 6 degrees	R7 target and SCM-PRC-001 updated; model and drawing notes state plus or minus 6 degrees. With the larger cluster the grip force at the plus or minus 6 degree edge is still 111 N (was 107 N at 8 degrees on the old cluster), so R6 remains not met; see item 14
13	Follow-on targets after the rework (R3, R5, R12)	Revisit once the parts are priced, as recommended. With the parts priced at TRL 3: R3 set to 17 steps/min on the present 250 W motor (was 20 steps/min; 20 would need 280 W); R5 set to 27 kg (was 25 kg); R12 and <code>budget_usd</code> set to \$650 (were \$600)	<code>project.yaml</code> <code>budget_usd</code> 600 to 650; SCM-REQ-001 R3, R5 and R12; SCM-CAL-001 v0.2 constants. Results: 238 W peak on the 250 W motor (5 % margin), 26.3 kg truck (0.7 kg margin), \$633 BOM (2.6 % margin). The figures follow from the priced parts; Amish may adjust them at any time

The pitch and the problem line in `project.yaml` are unchanged by these decisions.

Table 2. Items left open by this record, all decided on 2026-10-02.

#	Item	Status
9	First user group for interviews: gig couriers, a parcel company pilot, or appliance and moving crews	Decided by Amish, 2026-10-02: interview parcel couriers who deliver to walk-up apartment buildings first, with gig couriers as a second group; appliance and moving crews are left out (SCM-DEC-001, item 6)
10	Partner courier group or co-design partner	Decided by Amish, 2026-10-02: seek a regional parcel or last-mile delivery company willing to run a small supervised pilot, chosen from the interviews of item 6; the first candidate type to approach, not yet agreed (SCM-DEC-001, item 7)
14	New: grip force after the rework (R6). The 150 mm arm widens the swing of the center of mass about the support wheel from plus or minus 71 to 89 mm, so the grip force is 70 N at the set angle and 111 N at the plus or minus 6 degree window edge; plus or minus 4.4 degrees would be needed for 100 N. Options: (A) keep plus or minus 6 degrees and relax R6 to 115 N until grip force is measured at TRL 4; (B) tighten the window to plus or minus 4 degrees, which risks frequent stops on uneven stairs; (C) lengthen the handle within the 1,500 mm height limit, which gains only a few percent. Recommendation: A	Decided by Amish, 2026-10-02 (SCM-DEC-001, item 4), not option A: R6 stays at 100 N and the tilt window is tightened to plus or minus 3 degrees for the first loaded trials, and widened toward plus or minus 6 degrees only when grip force is measured at 100 N or less, or operators are shown to handle the measured force safely

■ Consequences

- R2 and R15 move from not met to met; R6 stays not met. On paper StepClimber now meets 10 of 15 requirements, misses 1 and leaves 4 that need hardware.
- Endurance falls from 1,394 to 1,327 loaded steps per charge because of the heavier truck and the second chain stage; R4 is still met.
- No cross-repo action arises: StepClimber keeps its own 24 V pack (SCM-DDR-001 item 6).
- TRL 3 is the hard stop. Nothing in this record starts TRL 4 work.